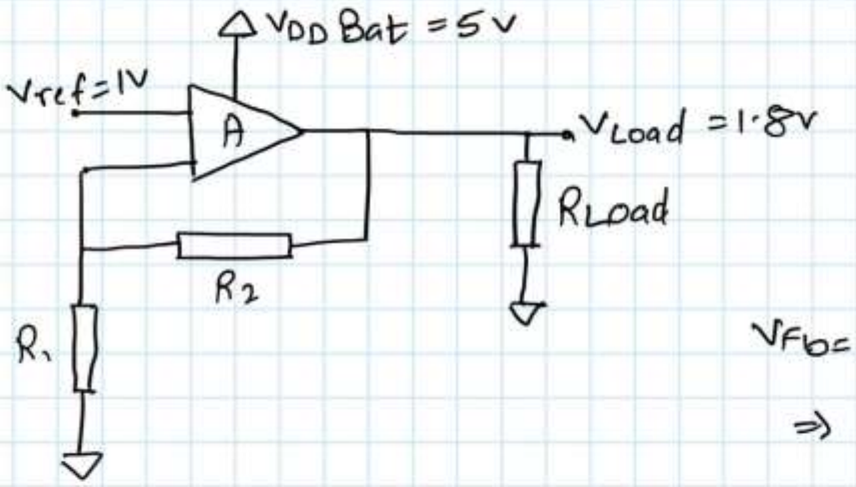
ET4382 - SMPC Mini-Project 1

Raghavendra Joshi 6438180
Daniel Tyukov 5714699

Part A

Part-1:



$V_{FB} = 1$ when $V_{load} = 1.8$

$$\Rightarrow \frac{1}{1.8} = \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow \frac{R_2}{R_1} = 0.8$$

So, lets assume $R_1 = 10\,K$; $R_2 = 8\,K$

$$\eta = \frac{V_{out}}{V_{in}} \quad [\text{Assuming } I_{out} = I_{in}]$$

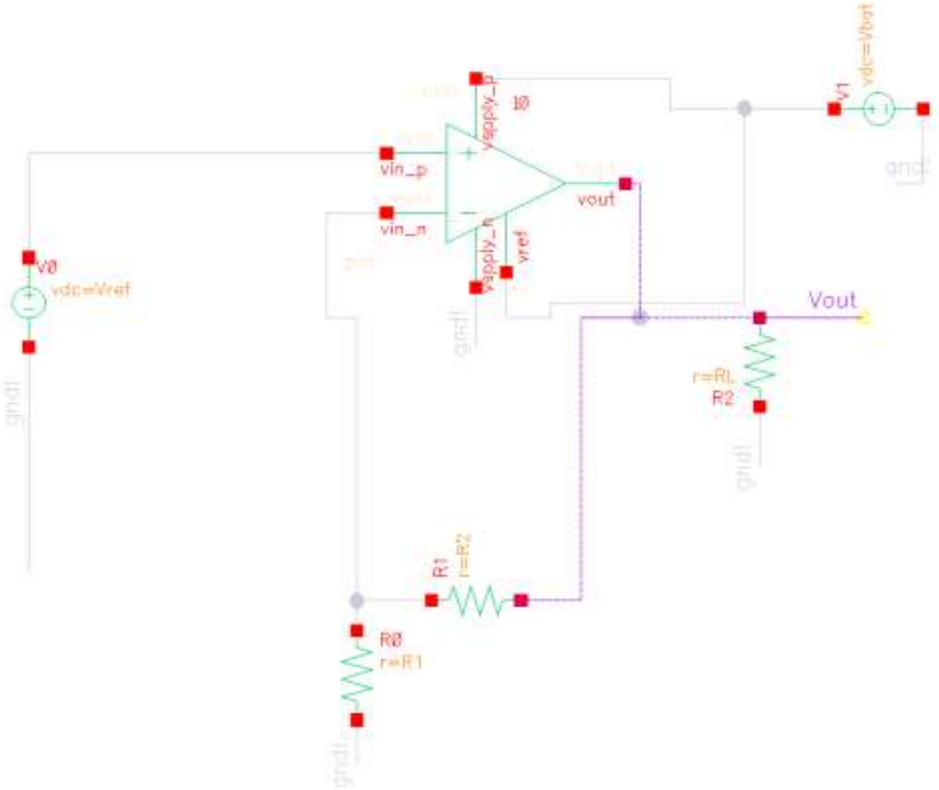
$$= \frac{1.8}{5}$$

$\Rightarrow \boxed{\eta = 0.36} \Rightarrow$ Quite low ☹

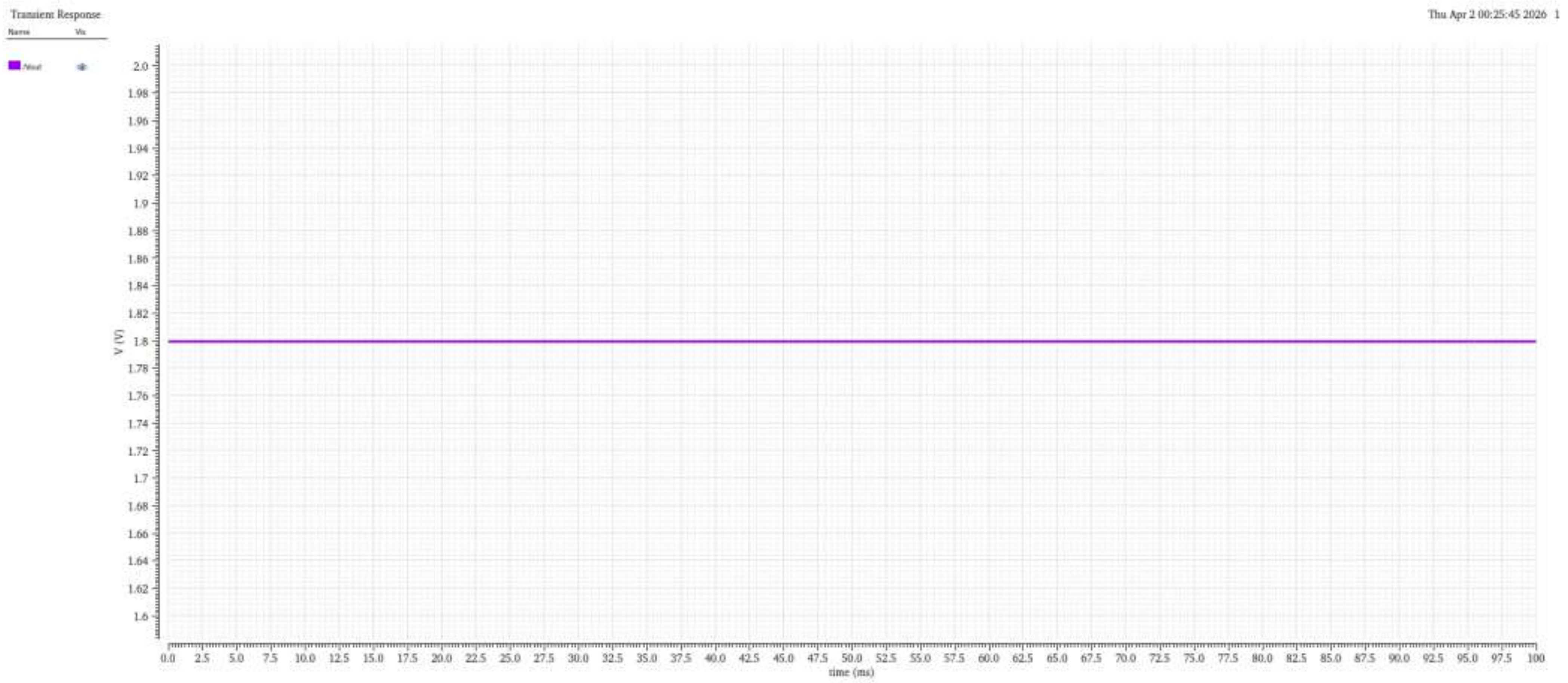
So is voltage regulator a good choice?

- NO!
- Efficiency =0.36
- That’s why LDOs have a very low voltage drop – to keep this efficiency high

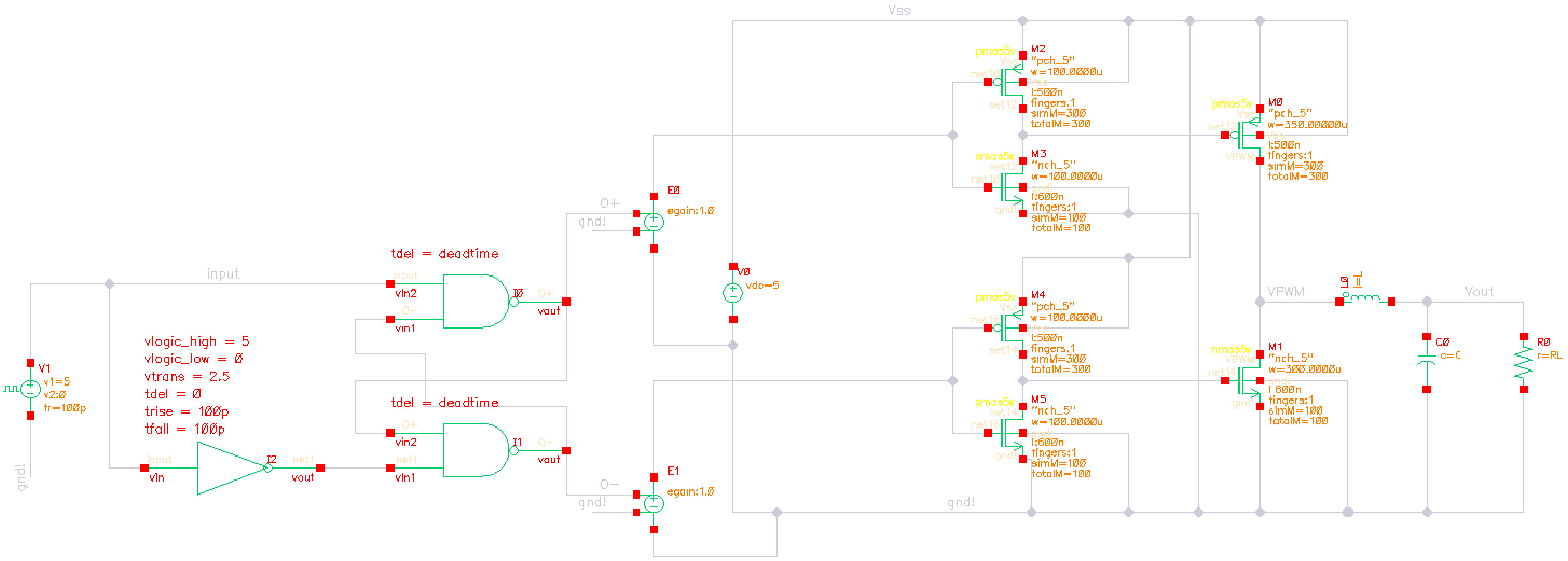
Schematic for Part A



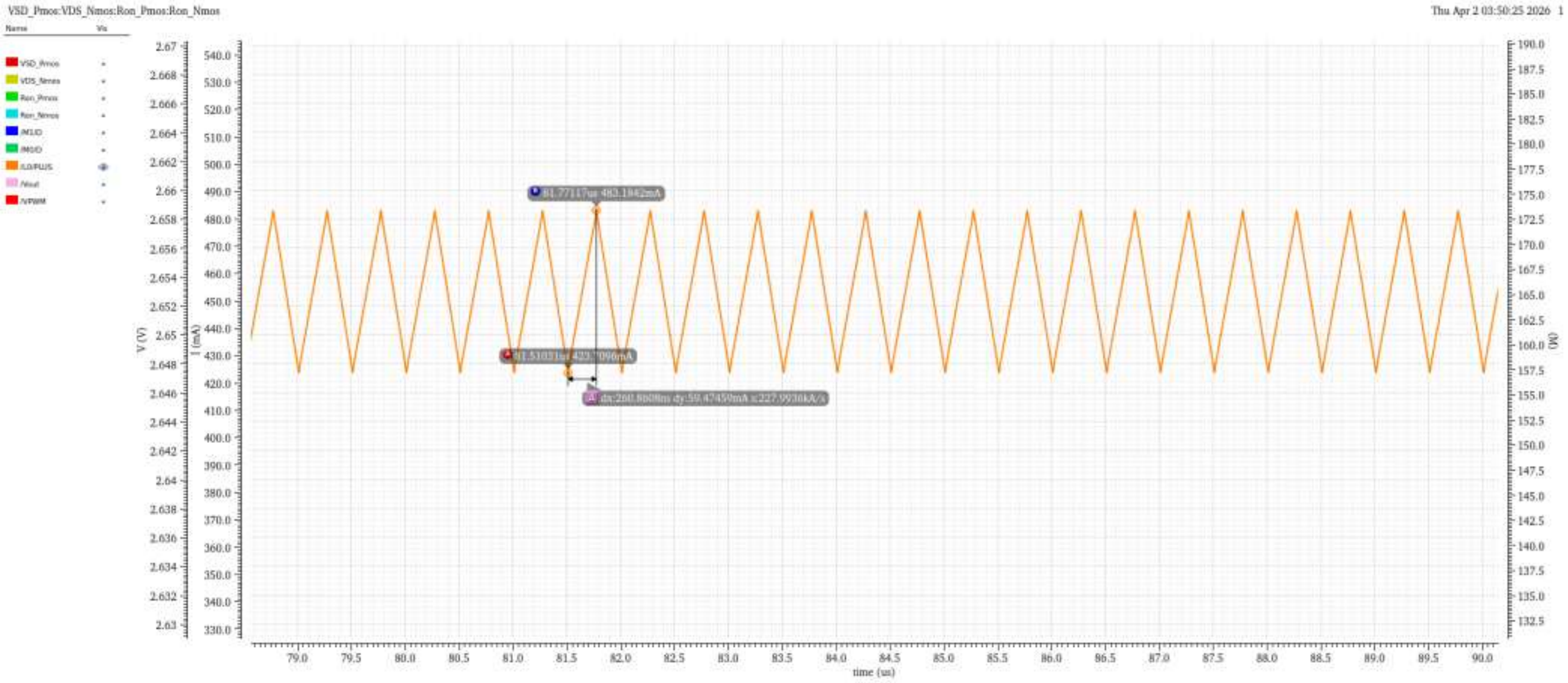
Vout plot for Part A



Part B - Schematic

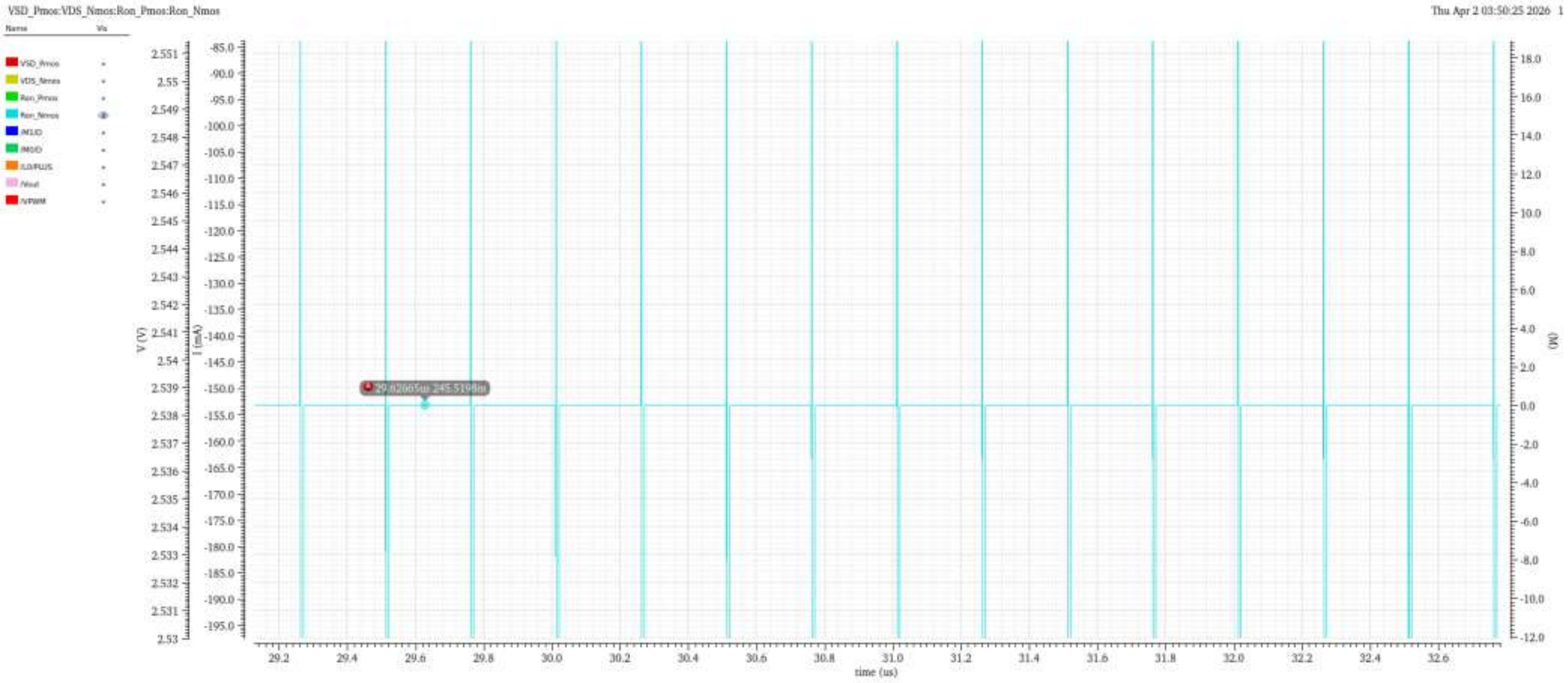


Iripple

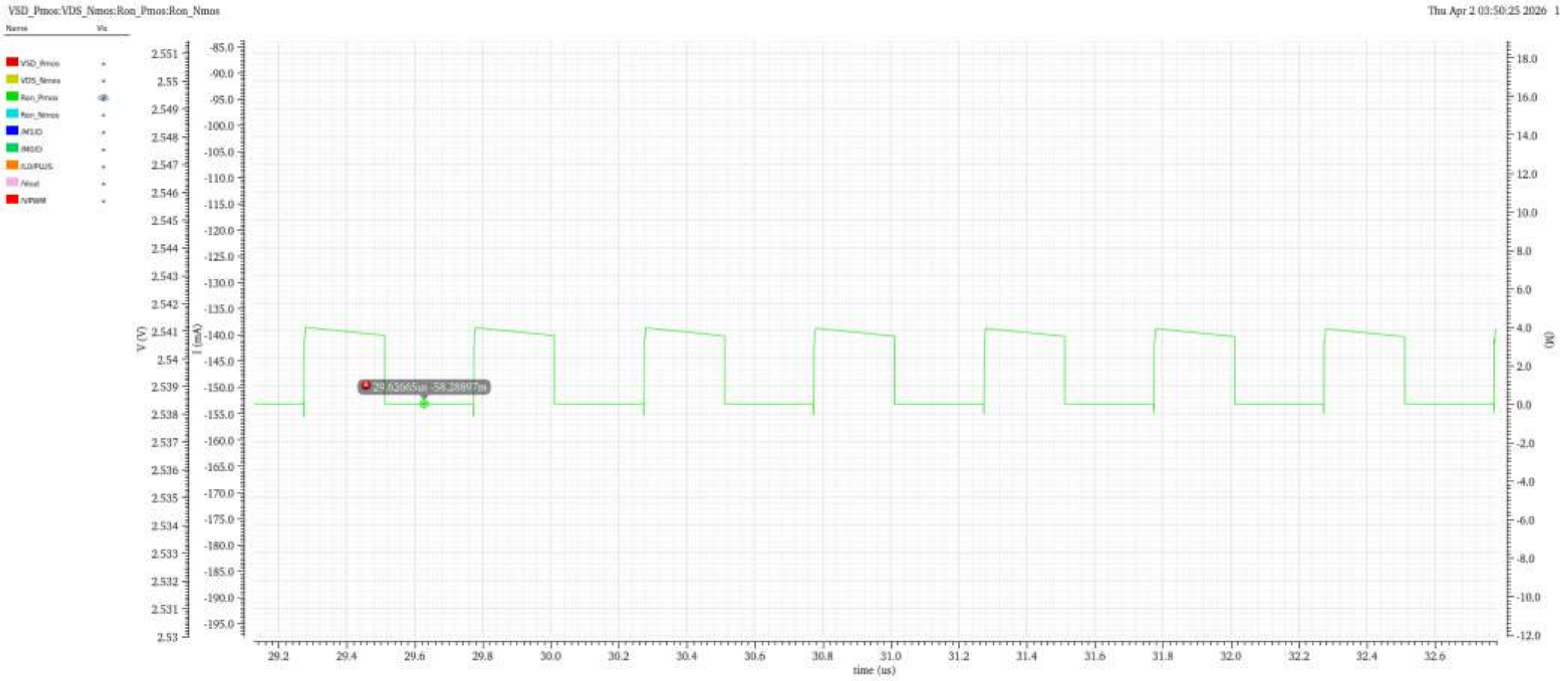


VSD_Pmos:VDS_Nmos:Ron_Pmos:Ron_Nmos
Thu Apr 2 03:50:25 2026 1
VSD_Pmos
VDS_Nmos
Ron_Pmos
Ron_Nmos
/M1/D
/M0/D
/L0/PLUS
/Vout
/VPWM
81.77117us 483.1842mA
81.51031us 423.7096mA
dx:260.8608ns dy:59.47459mA s:227.9936kA/s
V (V)
I (mA)
time (us)

Ron-Nmos


VSD_Pmos:VDS_Nmos:Ron_Pmos:Ron_Nmos
Thu Apr 2 03:50:25 2026 1
Name
VSD_Pmos
VDS_Nmos
Ron_Pmos
Ron_Nmos
/VPWM
V (V)
I (mA)
time (us)

Ron-Pmos



VSD_Pmos:VDS_Nmos:Ron_Pmos:Ron_Nmos
Thu Apr 2 03:50:25 2026 1
Name
Vis
VSD_Pmos
VDS_Nmos
Ron_Pmos
Ron_Nmos
/M1/D
/M0/D
/L0/PLUS
/Vout
/VPWM
V (V)
I (mA)
(M)
time (us)

Ron observations

- As I increased the W/L ratios of both the NMOS and PMOS, the extracted on-resistances increased in my simulations.
- The NMOS on-resistance was more sensitive to changes in W/L than the PMOS on-resistance.
- Based on this sizing study, the final device dimensions were chosen as:
  - PMOS: $W/L = 350\,\mu m/500\,nm$
  - NMOS: $W/L = 300\,\mu m/600\,nm$

Duty Cycle and Ripple Current Calculations

① Duty cycle $= \frac{V_o}{V_i} = \frac{1.8}{5} = 0.36$

at 2MHz, $T = \frac{1}{2\times10^6} = 500ns \Rightarrow t_{on} = DT = 180ns$

$\Rightarrow t_{off} = 320ns$

② Ripple current calc:

on-time: $V_L = V_{in} - V_o = 5 - 1.8 = 3.2V$

off-time: $V_L = -V_o = -1.8V$

So, (on-time) $\Delta I_L = \frac{(V_{in} - V_o)t_{on}}{L} = \frac{3.2 \times 180n}{10\mu} = 57.6mA$

(off-time) $\Delta I_L = \frac{(-1.8) \times 320n}{10\mu} = -57.6mA$

Ripple current $= 57.6mA$

$I_{avg} = \frac{V_o}{R_L} = \frac{1.8}{4} = 0.45A$

Pcond Loss Calculations (with obtained Ron and I values)

$$P_{cond} = P_{avg} + P_{ripple}$$

$$P_{avg} = I^2_{avg} R_{on,pmos} \, Duty + I^2_{avg} R_{on,nmos} (1 - Duty)$$

$$= (455.8m)^2 (58.28m)(0.36) + (455.8m)^2 (245.4m)(0.64)$$

$$= 0.037$$

$$P_{ripple} = I^2_{ripple} R_{on,pmos} \, Duty + I^2_{ripple} R_{on,nmos} (1 - Duty)$$

$$= (59.47m)^2 (58.28m)(0.36) + (59.47m)^2 (245.4m)(0.64)$$

$$= 6.3 \times 10^{-4}$$

$$P_{cond} = P_{avg} + P_{ripple} = 0.0376 \cong 0.038 \, W$$

$$\text{total power} = I(L_{avg}) \times V_L$$

$$= 455.8m \times 1.8 = 0.82 \, W$$

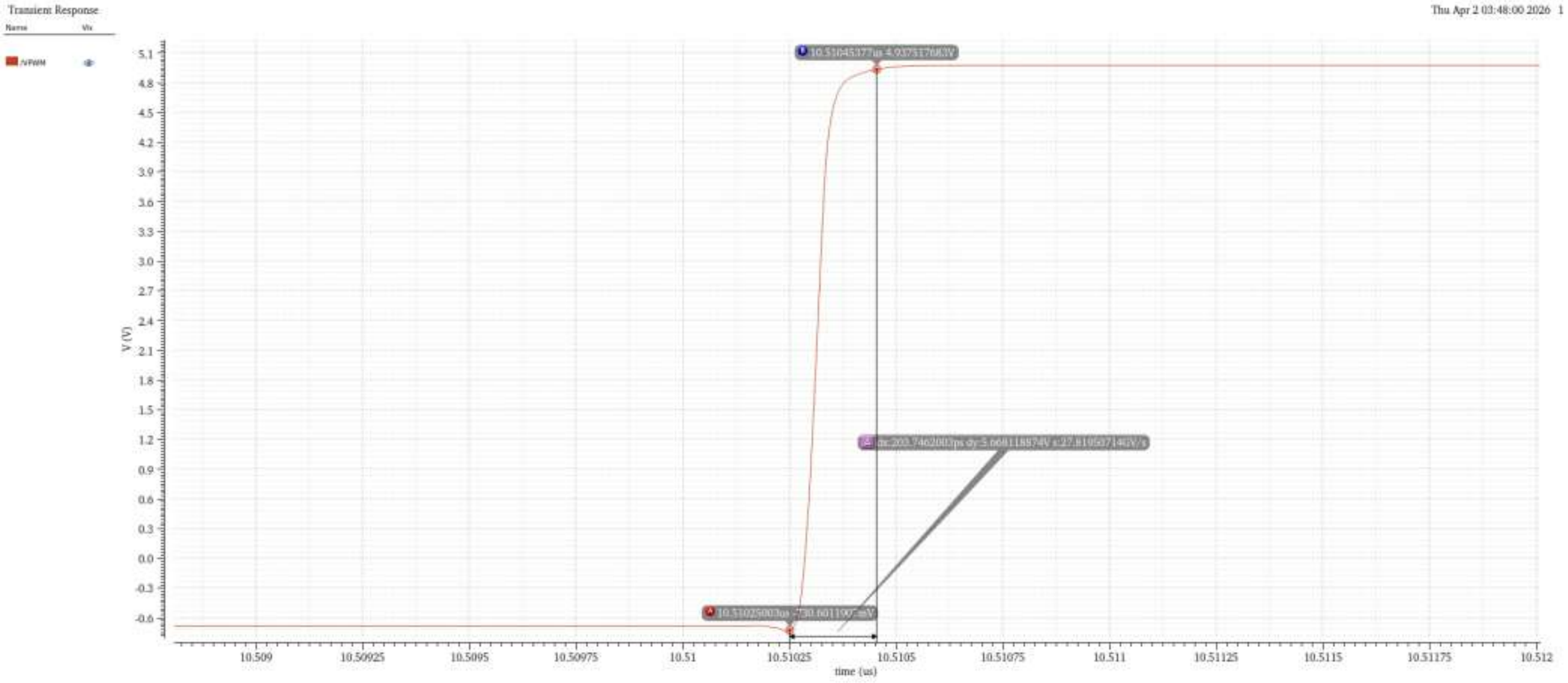
$$\frac{P_{cond}}{P_{total}} = \frac{0.038}{0.82} \cong 4.63\% \text{ of total power}$$

$$\boxed{\eta = 0.954}$$

Observations

- Pcond is < 5% of the total power
- Iavg matches in simulations and calculations

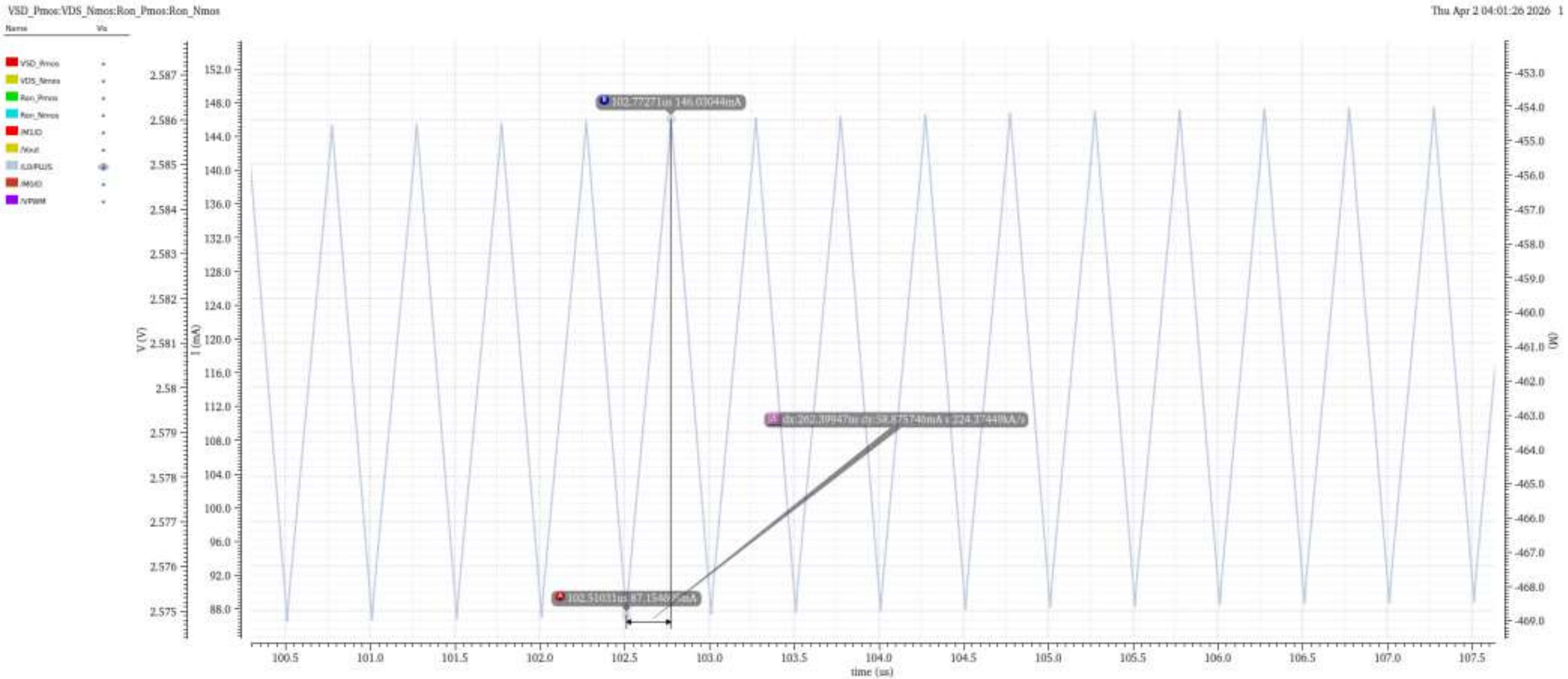
Transition time plot of VPWM



Gate Driver sizes

- Pmos: 100u/500n
- Nmos: 100u/600n
- Transition time < 2.0ns

16ohm load case (Ron values stay same, I plots change)



16ohm load case

5) 16 Ω conduction loss:

$P_{cond} = P_{avg} + P_{ripple}$

$P_{avg} = I^2_{avg} R_{on,pmos} Duty + I^2_{avg} R_{on,nmos} (1-Duty)$

$= (123.8m)^2 (58.28m)(0.36) + (123.8m)^2 (245.4m)(0.64)$

$= 2.73 \times 10^{-3} W$

$P_{ripple} = I^2_{ripple} R_{on,pmos} Duty + I^2_{ripple} R_{on,nmos} (1-Duty)$

$= (58.79m)^2 (58.28m)(0.36) + (58.79m)^2 (245.4m)(0.64)$

$= 6.15 \times 10^{-4} W$

$P_{cond} = P_{avg} + P_{ripple} = 3.345 \times 10^{-3} W$

Loss due to average current is dominant.

$\eta = \frac{P_{tot} - P_{cond}}{P_{tot}}$; $P_{tot} = I_{avg} \times 1.8 = 0.223 W$

$\boxed{\eta = 0.98}$ (increased efficiency as load increases)